CIE International A Level Physics

# Topic 17

# Oscillations

Revision Booklet

**This booklet covers:**

- Simple Harmonic Motion
- Displacement, Velocity & Acceleration
- Energy in SHM
- Damped Oscillations
- Forced Oscillations & Resonance

## Core Concepts and Definitions

### Oscillatory Motion — Key Terms

- **Displacement** $x$: the distance from the equilibrium position, with direction (m).
- **Amplitude** $x_0$: the maximum displacement from equilibrium (m).
- **Period** $T$: the time for one complete oscillation (s).
- **Frequency** $f$: the number of complete oscillations per second (Hz); $f = 1/T$.
- **Angular frequency** $\omega$: $\omega = 2\pi f = 2\pi/T$ (rad s$^{-1}$).
- **Phase difference** $\phi$: the fraction of a cycle by which one oscillation leads or lags another, expressed in radians.

### Simple Harmonic Motion (SHM)

An oscillation is **simple harmonic** if the acceleration of the object is:

- **proportional** to its displacement from a fixed equilibrium point, and
- always directed **towards** that equilibrium point (opposite to displacement).

$$a \propto -x$$

## Equations of Simple Harmonic Motion

### Defining Equation of SHM

$$a = -\omega^2 x$$

$a$ = acceleration (m s$^{-2}$)
$\omega$ = angular frequency (rad s$^{-1}$)
$x$ = displacement from equilibrium (m)

The negative sign confirms acceleration is always directed **opposite** to displacement.

### Displacement; Velocity and Acceleration

$$x = x_0 \sin \omega t \qquad \text{(starting from equilibrium)}$$

$$v = v_0 \cos \omega t \qquad \text{where } v_0 = \omega x_0$$

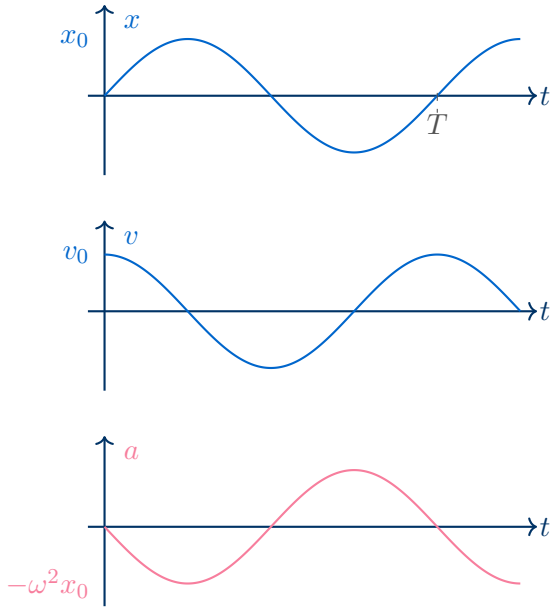
$$v = \pm\, \omega \sqrt{x_0^2 - x^2}$$

$x_0$ = amplitude (m)
$v_0$ = maximum speed = $\omega x_0$ (m s$^{-1}$)
$t$ = time (s)

## Maxima and Minima

- Acceleration is **maximum** at maximum displacement ($x = \pm x_0$): $|a_{\text{max}}| = \omega^2 x_0$
- Speed is **maximum** at the equilibrium position ($x = 0$): $v_{\text{max}} = \omega x_0$
- Speed is **zero** at the turning points ($x = \pm x_0$).

### Graphs of $x$, $v$ and $a$ against time




## Phase Relationships

- $v$ leads $x$ by $\frac{\pi}{2}$ rad (quarter period ahead).
- $a$ leads $v$ by $\frac{\pi}{2}$ rad, so $a$ is **antiphase** ($\pi$ rad ahead) with $x$.

## Common Mistake

The equation $x = x_0 \sin \omega t$ assumes the object starts at the **equilibrium position** at $t = 0$. If it starts at maximum displacement, use $x = x_0 \cos \omega t$ instead. Always check the initial conditions.

## Energy in Simple Harmonic Motion

### Total Energy of a SHM System

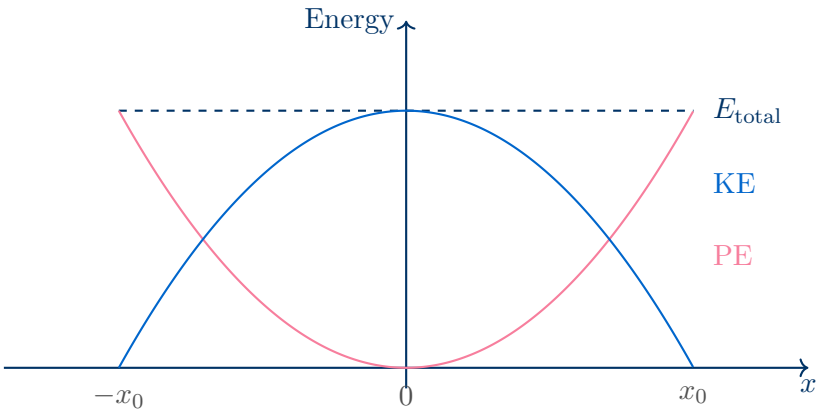
$$E = \frac{1}{2}m\omega^2 x_0^2$$

$E$ = total mechanical energy (J)
$m$ = mass of the oscillating object (kg)
$\omega$ = angular frequency (rad s$^{-1}$)
$x_0$ = amplitude (m)

### Energy Interchange During SHM

- At the **equilibrium position** ($x = 0$): KE is maximum, PE is zero.
- At the **turning points** ($x = \pm x_0$): KE is zero, PE is maximum.
- The **total energy remains constant** throughout (assuming no damping).
- KE $= \frac{1}{2}m\omega^2(x_0^2 - x^2)$ PE $= \frac{1}{2}m\omega^2 x^2$

**Energy–displacement graph**




## Damped and Forced Oscillations

### Damping

#### Damping

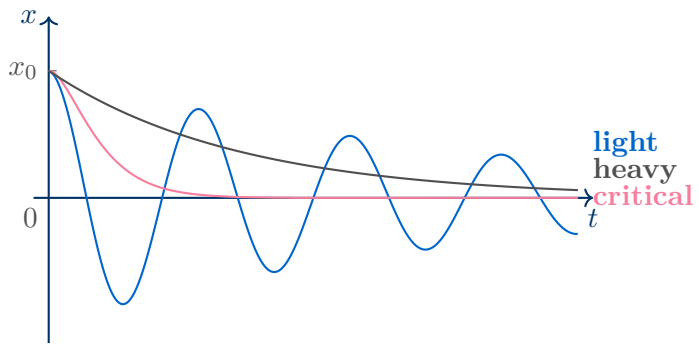
**Damping** occurs when a resistive force (e.g., air resistance, viscosity) acts on an oscillating system. Energy is removed from the system, causing the amplitude to **decrease over time**. The frequency is approximately unchanged for light damping.

#### Types of Damping

- **Light damping**: amplitude decreases gradually over many oscillations; the system oscillates for a long time before coming to rest.

- **Critical damping**: the system returns to equilibrium in the **shortest possible time** without oscillating. Used in car suspension and door closers.
- **Heavy (overdamping)**: the system returns to equilibrium **slowly** without oscillating; slower return than critical damping.

**Displacement–time graphs for the three types of damping**




## Forced Oscillations and Resonance

### Forced Oscillations and Natural Frequency

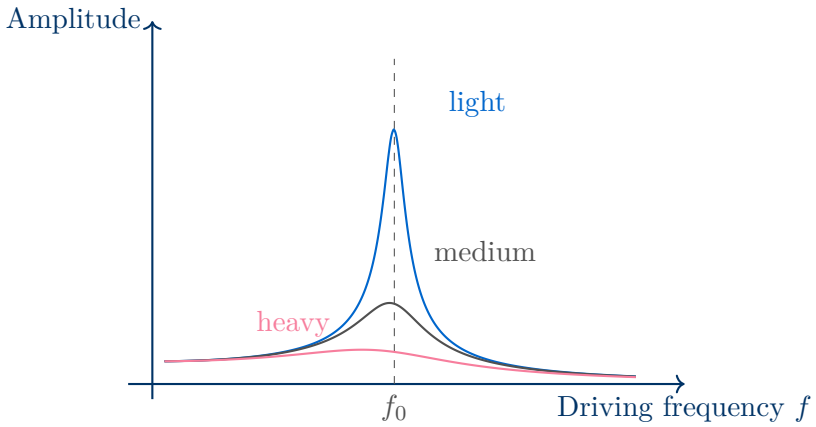
- Every oscillating system has a **natural frequency** $f_0$ at which it oscillates freely when displaced and released.
- A **forced oscillation** occurs when a periodic **driving force** is applied to the system at a **driving frequency** $f$.
- The system oscillates at the **driving frequency**, not necessarily at its natural frequency.

### Resonance

**Resonance** occurs when the driving frequency equals the natural frequency of the system ($f = f_0$). At resonance:

- The amplitude of oscillation is a **maximum**.
- Energy transfer from the driver to the system is most efficient.
- The degree of damping determines how sharp the resonance peak is and the maximum amplitude reached.

### Amplitude–frequency graph (resonance curves)




**Effect of Damping on Resonance**

- More damping ⇒ lower and broader resonance peak.
- More damping ⇒ peak shifts **below** $f_0$ (towards lower frequencies).
- Less damping ⇒ sharper, taller peak with maximum amplitude closer to $f_0$.
- With no damping the peak would be infinite at exactly $f_0$.

**Resonance in Real Life**

Resonance can be **useful** (e.g. MRI scanners, musical instruments, microwave ovens) or **destructive** (e.g. bridges vibrating in wind, buildings in earthquakes). Engineers use damping to control unwanted resonance.

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $a = -\omega^2 x$ | SHM defining equation | m s$^{-2}$ |
| $\omega = 2\pi f = \dfrac{2\pi}{T}$ | Angular frequency | rad s$^{-1}$ |
| $x = x_0 \sin \omega t$ | Displacement (from equilibrium at $t = 0$) | m |
| $v = v_0 \cos \omega t$ | Velocity | m s$^{-1}$ |
| $v = \pm\omega\sqrt{x_0^2 - x^2}$ | Speed at displacement $x$ | m s$^{-1}$ |
| $v_0 = \omega x_0$ | Maximum speed | m s$^{-1}$ |
| $a_{\text{max}} = \omega^2 x_0$ | Maximum acceleration | m s$^{-2}$ |
| $E = \frac{1}{2}m\omega^2 x_0^2$ | Total energy of SHM system | J |
| $\text{KE} = \frac{1}{2}m\omega^2(x_0^2 - x^2)$ | Kinetic energy at displacement $x$ | J |
| $\text{PE} = \frac{1}{2}m\omega^2 x^2$ | Potential energy at displacement $x$ | J |

**Key relationships:** $T = 1/f$, $\omega = 2\pi f$, $v_{\text{max}} = \omega x_0$, $a_{\text{max}} = \omega^2 x_0$
**Phase:** $v$ leads $x$ by $\pi/2$; $a$ is antiphase with $x$ (leads by $\pi$)

## Worked Examples

### Example 1 — Finding Angular Frequency and Max Speed

**Question:** A mass oscillates with SHM, amplitude 4.0 cm and period 0.80 s. Calculate (a) the angular frequency and (b) the maximum speed.

**Solution**

**(a)** $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{0.80} = \mathbf{7.85}$ rad s$^{-1}$

**(b)** $v_{\text{max}} = \omega x_0 = 7.85 \times 0.040 = \mathbf{0.31}$ m s$^{-1}$

### Example 2 — Speed at a Given Displacement

**Question:** The same mass ($\omega = 7.85$ rad s$^{-1}$, $x_0 = 4.0$ cm) has displacement $x = 2.5$ cm. Find its speed.

**Solution**

$$v = \omega\sqrt{x_0^2 - x^2} = 7.85 \times \sqrt{(0.040)^2 - (0.025)^2}$$

$$v = 7.85 \times \sqrt{1.60 \times 10^{-3} - 6.25 \times 10^{-4}} = 7.85 \times \sqrt{9.75 \times 10^{-4}}$$

$$v = 7.85 \times 0.0312 = \mathbf{0.245}\ \text{m s}^{-\mathbf{1}}$$

### Example 3 — Total Energy of SHM System

**Question:** A 0.20 kg mass oscillates with $\omega = 7.85$ rad s$^{-1}$ and amplitude 4.0 cm. Calculate the total energy.

**Solution**

$$E = \tfrac{1}{2}m\omega^2 x_0^2 = \tfrac{1}{2} \times 0.20 \times (7.85)^2 \times (0.040)^2$$

$$E = 0.10 \times 61.6 \times 1.60 \times 10^{-3} = \mathbf{9.9 \times 10^{-3}}\ \text{J}$$

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** State the two conditions that define simple harmonic motion.

*[2 marks]*

**Q2.** Define the terms *amplitude*, *period* and *angular frequency* for an oscillating system.

*[3 marks]*

**Q3.** An object undergoes SHM with frequency 2.5 Hz and amplitude 6.0 cm.

(a) Calculate the angular frequency. *[1 mark]*

(b) Calculate the maximum acceleration. *[2 marks]*

(c) Calculate the speed when $x = 4.0$ cm. *[2 marks]*

**Q4.** Sketch displacement–time graphs for *light*, *critical* and *heavy* damping on the same axes, starting from the same initial displacement. Label each curve.

*[3 marks]*

**Q5.** Explain what is meant by *resonance* and state the condition under which it occurs.

*[2 marks]*

## Section B — Longer Structured Questions

**Q6.** A 0.15 kg mass is attached to a spring and oscillates vertically with SHM. Its displacement is given by $x = 0.050\sin(12t)$, where $x$ is in metres and $t$ in seconds.

(a) Write down the amplitude and angular frequency of the motion.

*[2 marks]*

(b) Show that the maximum speed is 0.60 m $s^{-1}$.

*[2 marks]*

(c) Calculate the total energy of the oscillation.

*[2 marks]*

(d) Sketch graphs on the same axes showing how the kinetic energy and potential energy vary with displacement $x$. Label your axes clearly.

*[3 marks]*

**Q7.** A child on a swing is pushed periodically by a parent.

(a) Explain why the amplitude of the swing increases when the pushing frequency equals the natural frequency of the swing.

*[2 marks]*

(b) The parent now pushes at a frequency higher than the natural frequency. Describe and explain what happens to the amplitude.

*[2 marks]*

(c) Air resistance acts on the swing. Describe how this affects the resonance curve compared to an undamped system.

*[2 marks]*

## Mark Scheme and Answers

**Q1.** Acceleration is proportional to displacement from a fixed (equilibrium) point [1]; acceleration is always directed towards that point / opposite to displacement [1].

**Q2.** Amplitude: maximum displacement from equilibrium [1]. Period: time for one complete oscillation [1]. Angular frequency: $\omega = 2\pi f = 2\pi/T$ (rad s$^{-1}$) [1].

**Q3(a).** $\omega = 2\pi f = 2\pi \times 2.5 = \mathbf{15.7}$ rad s$^{-1}$ [1].

**Q3(b).** $a_{\max} = \omega^2 x_0 = (15.7)^2 \times 0.060$ [1] $= \mathbf{14.8}$ m s$^{-2}$ [1].

**Q3(c).** $v = \omega\sqrt{x_0^2 - x^2} = 15.7 \times \sqrt{(0.060)^2 - (0.040)^2}$ [1] $= 15.7 \times 0.0447 = \mathbf{0.70}$ m s$^{-1}$ [1].

**Q4.** All three start from same $x_0$ [1]; light damping: decaying sinusoid, multiple oscillations; critical: returns to zero smoothly in shortest time without oscillating; heavy: slower return to zero, no oscillation — all three correctly labelled [2].

**Q5.** Resonance: when the driving frequency equals the natural frequency of the system [1]; the amplitude of oscillation reaches a maximum [1].

**Q6(a).** Amplitude $x_0 = 0.050$ m [1]; angular frequency $\omega = 12$ rad s$^{-1}$ [1].

**Q6(b).** $v_{\text{max}} = \omega x_0 = 12 \times 0.050 = 0.60$ m s$^{-1}$ [2] (must show substitution).

**Q6(c).** $E = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2} \times 0.15 \times 144 \times 2.5 \times 10^{-3}$ [1] $= \mathbf{2.7 \times 10^{-2}}$ J [1].

**Q6(d).** PE: upward parabola, zero at $x = 0$, maximum $E$ at $x = \pm x_0$ [1]; KE: inverted parabola, maximum $E$ at $x = 0$, zero at $x = \pm x_0$ [1]; axes correctly labelled with $E$, $-x_0$, 0, $x_0$ [1].

**Q7(a).** Energy is transferred most efficiently from driver to system when $f_{\text{drive}} = f_0$ [1]; each push is in phase with the motion so energy is added each cycle, increasing amplitude [1].

**Q7(b).** Amplitude decreases [1]; driving frequency is not matched to natural frequency so energy transfer is less efficient / pushes are out of phase with motion [1].

**Q7(c).** The resonance peak is lower (smaller maximum amplitude) [1]; the peak is broader and shifts slightly to a frequency below $f_0$ [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define displacement, amplitude, period, frequency and angular frequency | |
| ☐ | State and apply the two conditions for SHM | |
| ☐ | Use $a = -\omega^2 x$ to identify and analyse SHM | |
| ☐ | Use $x = x_0 \sin \omega t$ and $v = v_0 \cos \omega t$ | |
| ☐ | Use $v = \pm\omega\sqrt{x_0^2 - x^2}$ to find speed at any displacement | |
| ☐ | Sketch and interpret graphs of $x$, $v$ and $a$ against $t$ | |
| ☐ | Describe the phase relationships between $x$, $v$ and $a$ | |
| ☐ | Describe the interchange between KE and PE during SHM | |
| ☐ | Use $E = \frac{1}{2}m\omega^2 x_0^2$ for total energy | |
| ☐ | Sketch and interpret KE and PE against displacement graphs | |
| ☐ | Explain and distinguish light, critical and heavy damping | |
| ☐ | Sketch displacement–time graphs for the three types of damping | |
| ☐ | Explain resonance and the condition for it to occur | |
| ☐ | Describe the effect of damping on the resonance curve | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

Remember: in SHM, acceleration and displacement are always linked by $a = -\omega^2 x$. If you can start from this equation and derive everything else, you truly understand the topic.